Evaluation of the use of LLM-based agents for social social simulation

Python ML LangChain LLM-agent Research

Introduction

The emergence of human-like behavior in Large Language Models (LLMs) has inspired various research efforts to evaluate LLM agents in studies of human behaviours. Specifically, LLM agents simulate humans to conduct user studies.

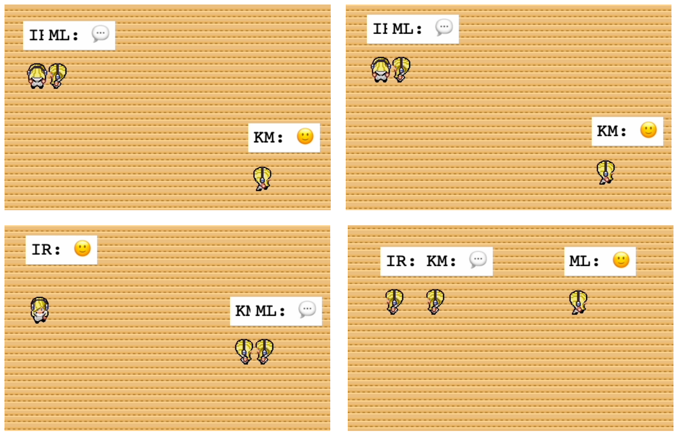
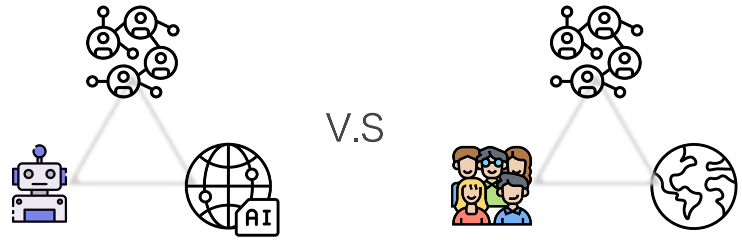


Fig. 2-1 LLM agents Simulated Visualization: Isabella, Klaus Mueller and Mackenzie were talking to each others

The problem I want to solve is:

- How reliable is using LLM agents to simulate human behavior?

In this work, I attempt to use social networks metrics as to investigate the reliability of LLM agents social simulation

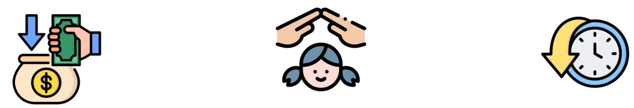


LLM agents generated social network

Real world social network

Contribution

- Practical Contribution: Compared to experiments involving human subjects, LLM-based social simulations



more affordable

eliminate the risk to human participants

less time-consuming

- When evaluating the simulation results, previous studies focus on overall network structure
- In contrast, we assess edge-level precision using real-world datasets to provide deeper insights into prediction reliability.

Method: Pure Simulation Approach

In this stage, we focus on evaluating the features of LLM network prediction behaviors in a purely simulated environment.
Our simulation analyzes the network characteristics of LLM agents, which are constructed based on their interaction records

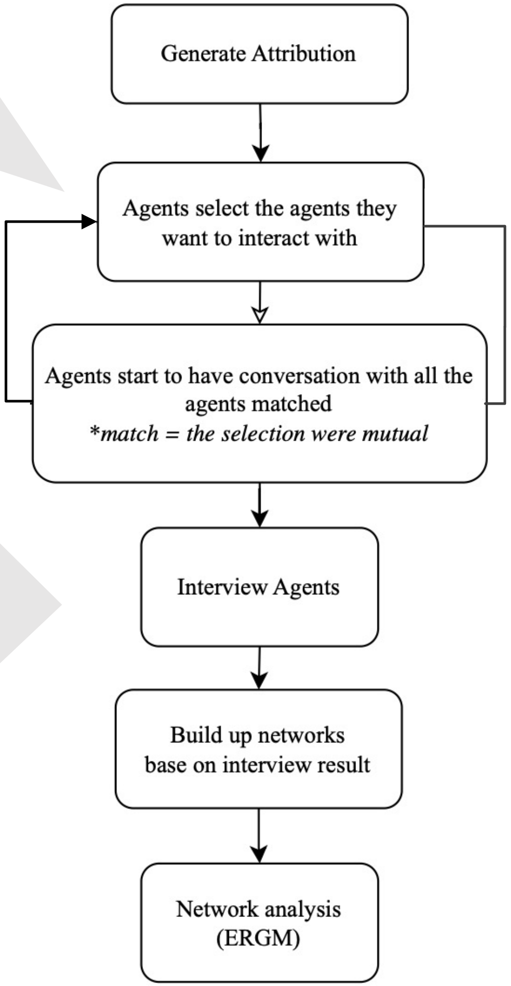
Fig. 2-2 Research Flow

Select Agent Prompt

🧑‍💻 Who do you most likely want to talk to based on the provided information?
{Agent attribution}
{self introductions}
{conversation history}
Your name is {name}. Don't select yourself.
Select the top 5
Respond in a comma separated list:
e.g. name1,name2,name3

Interview Prompt

{agent_summary_description}
It is {current_time}.
{agent_name}'s status: {currently}
Summary of relevant context from {agent_name}'s memory:
{relevant_memories}
Most recent observations:
{most_recent_memories}"
Observation: Interviewer said who are your best friends
🧑‍💻 Who do you consider as friends based on the previous conversations you have had?
Select at most 5 friends that are closest to you. Your name is {name}. Don't select yourself. Respond in a comma separated list:
e.g.'name1,name2,name3'"



Generate Name/Race/Gender/Age → Public Profiles

🧑‍💻: Provide a list of unique English names
🧠: ["Serenity Frost" , "Phoenix Shadow" , …]
🧑‍💻: Please finish the following sentence: My name is Serenity Frost. My gender identity is ____.
🧠: Female

Generate Big5 personality

🧑‍💻: Please finish the following sentence: I [1/2/3/4/5] I see myself as someone who is reserved

Generate self introduction base on attribution

Algorithm 1 Manage Conversations with Agents

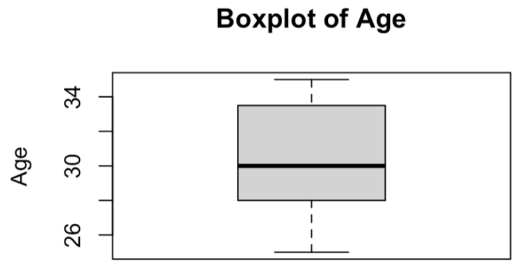
for *round* = 0 to *TotalRounds* **do**
  for *agent* = 0 to *TotalAgents* **do**
    if *round* = 0 **then**
      if Agent is Available **then**
        Select agent to start a conversation based on Self-Introduction
      else
        if Agent is in conversation **then**
          Select Continue Conversation or end Conversation
          if Agent is not in conversation **then**
            Select an agent to start a conversation based on memory

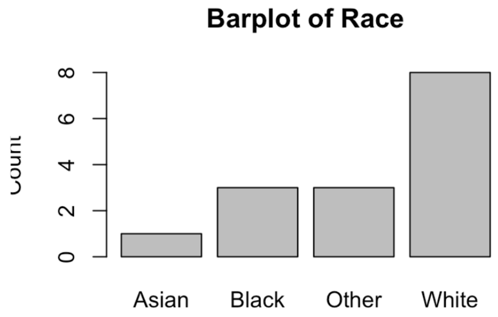
Fig. 2-3 Interaction mechanism design

Do factors affect the network formation between in human network also exist in LLM agent network?

Finding 1

Generated LLM attribution tend to be young, white and female, which cannot reflect the demographic features in real world





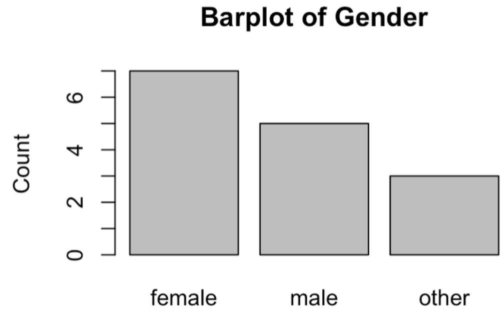


Fig. 2-4 LLM agents attribution distribution

Finding 2

Unlike human network, Gender/Race/Age is not a significant factor to affect the network formation

		Model 1
Intercept	edges	-3.82^{***}
		(0.50)
Test if tend to select those who have common friend	transitiveties	2.13^{***}
		(0.48)
Test if selections are mutual	mutual	-0.15
		(0.61)
Homophily variables: If significant, indicating that those who have same gender/race/age tend to connect to each other	nodematch.race	0.05
		(0.29)
	nodematch.gender	0.24
		(0.32)
	nodematch.age	0.30
		(0.42)
	AIC	192.78
	BIC	212.86
	Log Likelihood	-90.39

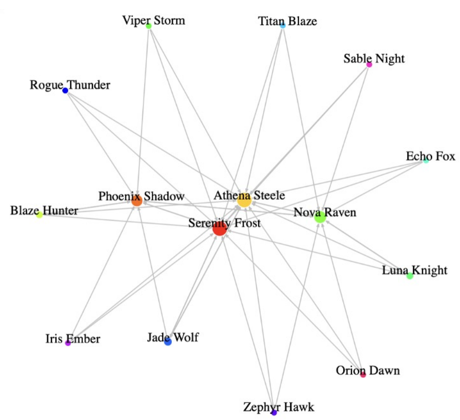
$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

Fig. 2-5 ERG Model

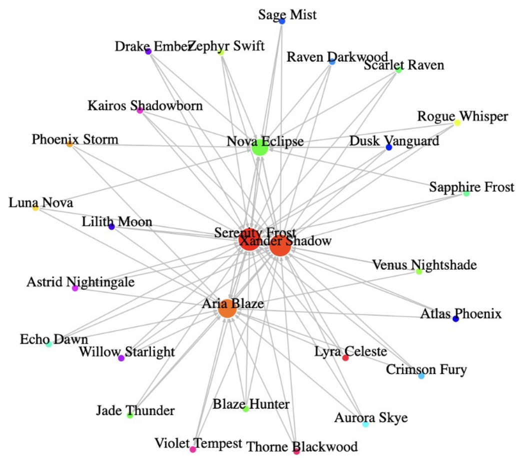
Finding 3

Networks developed by agents demonstrate highly centralized network structures

Fig. 2-1 LLM Agents Networks



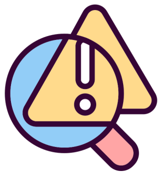
15 agents
10 rounds of conversations



30 agents
10 rounds of conversations

Who occupied the center points in the network?

The agents who appear at the top of the candidate agents list



Due to the natural of LLM agent memory retrieval system design, LLMs tend to prioritize the information they received first

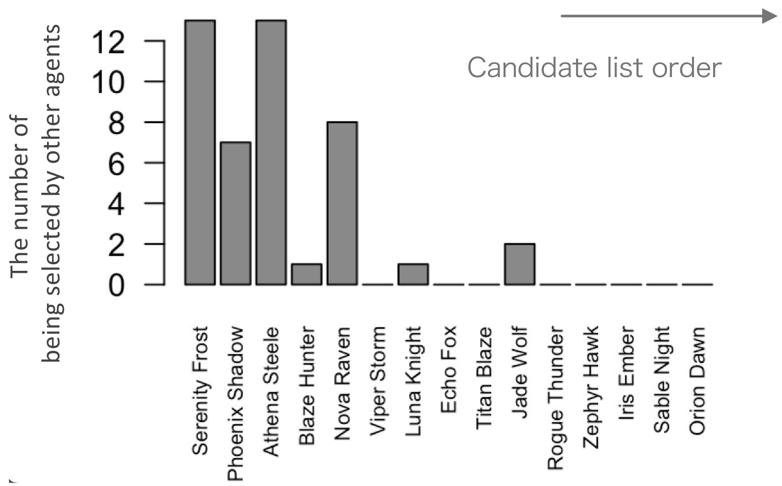


Fig. 2-6 LLM agents in degree distribution

Improved Simulation Method: Data Ground Approach

To overcome those limitation, I improved our method by conduct simulation base on a real world dataset.

Generated Agent Attributions

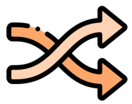
Generate Agent attributions base on the demographic distribution in dataset

Validate Result

With the real network dataset, instead of simply testing of the general network structure features, we could testing the precision of network tis predictions

Shuffle Input

Shuffle the order of input agent information when asking LLM agents to select their friend to minimize the effect of input orders



Method: Data Ground Approach

Step 0
Data

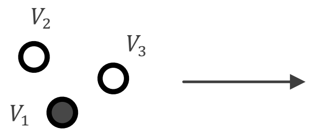
Indian village microfinance dataset [1]:
77 Indian villages.

Recorded the personal relationships of villagers(e.g. Who would you borrow money from? Ask for advice? … etc.)

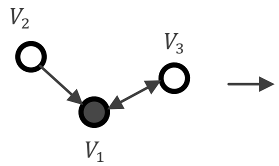
Recorded individual attributes: gender; caste, mother tongue ⋯ etc.)

Step 1
Data Preprocess

- Find all possible pair of connections between any two villagers: $(V_i,\ V_j)$



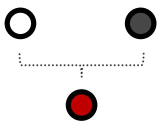
Survey:
V_3: I borrow money from V_2
V_2: I borrow money from V_1
V_3: I borrow rice from V_3



Edge(V_1, V_2)	X	Y
(1, 2)	$V_{1,gender,}, V_{2,gender}, \cdots$	0
(2, 1)	$V_{2,gender,}, V_{1,gender}, \cdots$	1
(1, 3)	$V_{1,gender,}, V_{3,gender}, \cdots$	1
(3, 1)	$V_{3,gender,}, V_{1,gender}, \cdots$	1
(2, 3)	$V_{2,gender,}, V_{3,gender}, \cdots$	0
(3, 2)	$V_{3,gender,}, V_{2,gender}, \cdots$	0

Step 2
Identified influential variables affect the edge formation

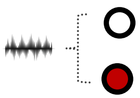
Since the number of not exist edges is much more the existed edge, we applied balance method to prevent bias prediction



New edge

SMOTE: Generate new edge (Y=1)

1. Selected a random edge
2. Find the sample’ s nearest nearest KNN neighbors in features space
3. Generate a new edge between them



New edge

ROSE: Generate new edge ((Y=1)

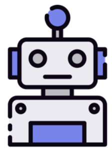
1. Selected a random edge
2. Create a small noise according to data distribution
3. Generate a news edge

Drop edge

Down Sampled: drop not existed edge (Y=0)

1. Selected a random not existed edge
2. Drop it

- **Machine Learning Base Lasso Logistics regress model**
  - Y= if the edge exist
  - X = demographic features of $V_i,\ \ V_j$



Build up LLM agents using influential variables as attributes!

[1] Banerjee, A., Chandrasekhar, A. G., Duflo, E., & Jackson, M. O. (2013). The diffusion of microfinance. Science, 341(6144), 1236498

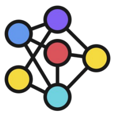
Step 3
Conduct Simulation: Asking LLM agent to predict network ties

[1] Chang, S., Chaszczewicz, A., Wang, E., Josifovska, M., Pierson, E., & Leskovec, J. (2024). LLMs generate structurally realistic social networks but overestimate political homophily. arXiv preprint arXiv:2408.16629.

Prompt

We applied global, local, and sequential methods, which have been validated in predicting network structures identical to general human network [1]

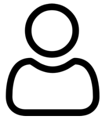
Global
Make entire network at once



You will be provided a list of people in the network. Please provide a list of friendship pairs in the format ID,

1. **Woman, Asian, 54, HINDUISM**
2. **Man, White, 18, HINDUISM**
3. **Woman, Black, 39, HINDUISM**

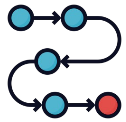
Local
Assign one persona at a time, no network info



You are **a Woman, age 23, HINDUISM,** You are joining a social network. Which of these people will you become friends with?

1. **Woman, Asian, 54, HINDUISM**
2. **Man, White, 18, HINDUISM**
3. **Woman, Black, 39, HINDUISM**

Sequential
Similar with the local method, but add the network information

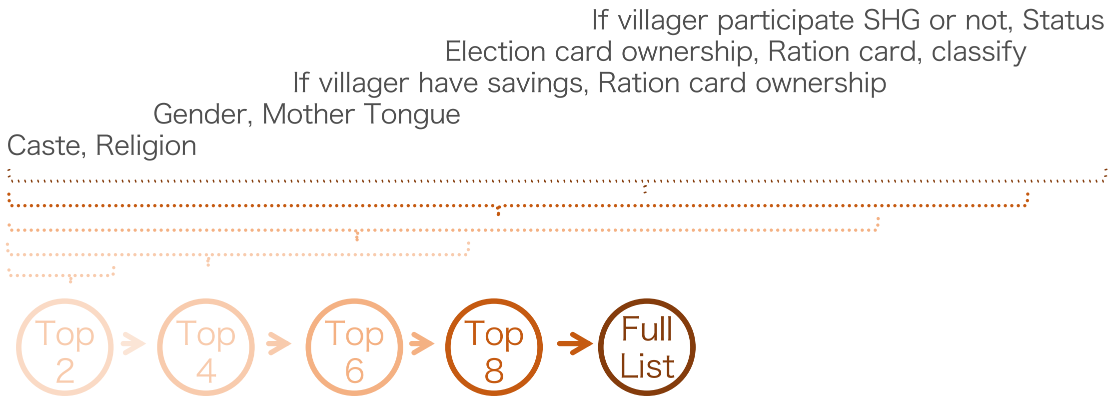


You are **a Woman, age 23, HINDUISM**, You are joining a social network. Which of these people will you become friends with?

1. **Woman, Asian, 54, HINDUISM, 12 friends**
2. **Man, White, 18, HINDUISM, 5 friends**
3. **Woman, Black, 39, HINDUISM, 16 friends**

The Number of Attributions

- Additionally, we explored whether using more attributes during simulation would enhance edge prediction performance in the network. We tested this by using the top 2, 4, 6, 8, and 11 attributes.



Results: Identifying Influential Attributions in Real Dataset

Balance Methods Evaluation

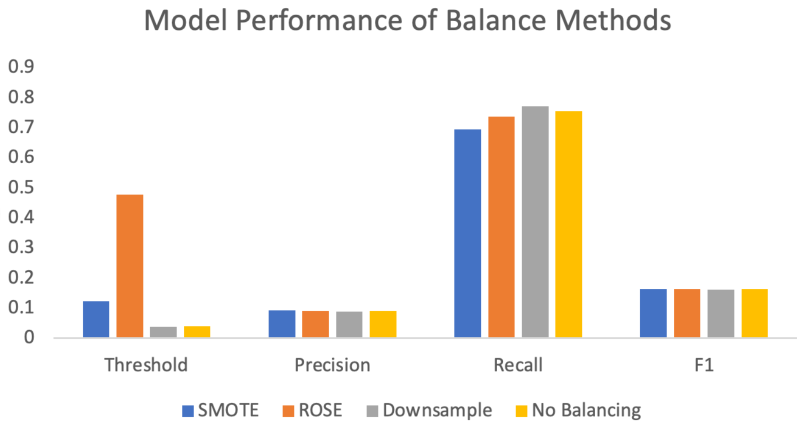


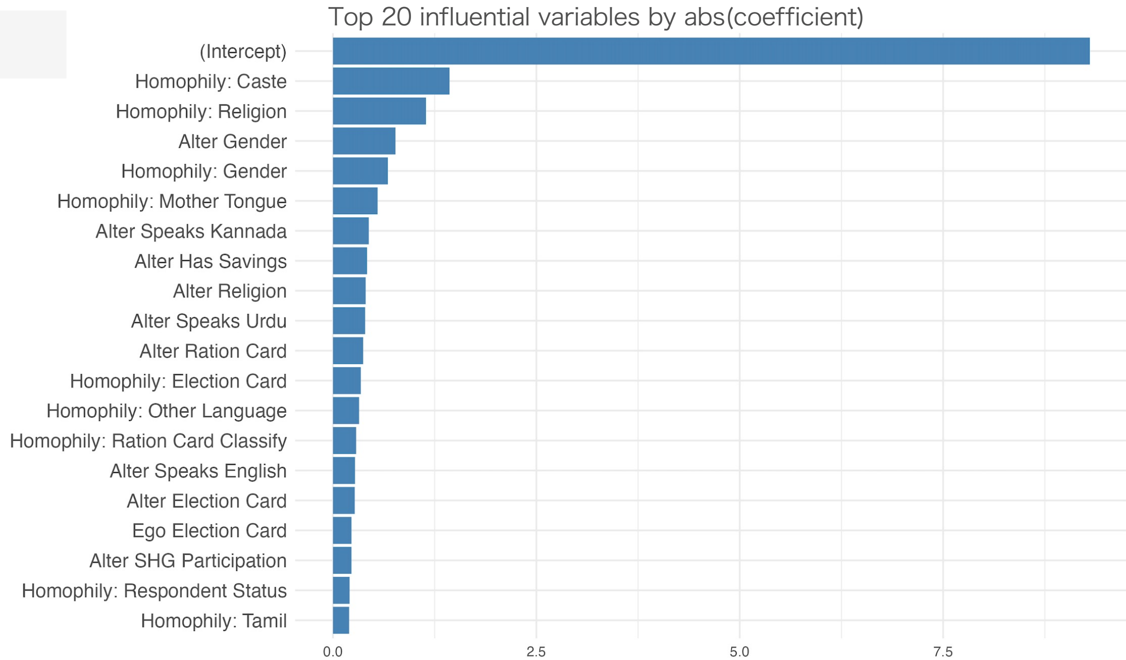
Fig. 2-7 Balance method comparsion

- All methods have similar precision, F1 and recall.
- SMOTE have highest overall F-1 score

Logistics Model Applied SMOTE on Predicting edges in the village networks

$* (V_i, V_j)$ = (Ego, Alter)

Fig. 2-8 Histogram of abs(coef)



Identify Influential Variables

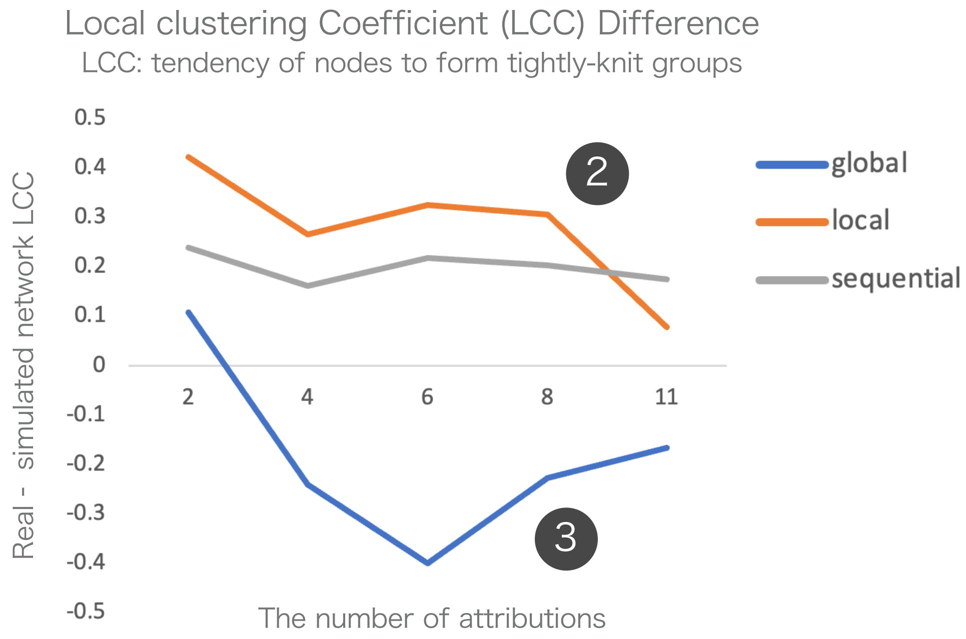
- Rank from the most influential variable to the least influential variables:
  1. Caste
  2. Religion
  3. Gender
  4. Mother
  5. Tongue
  6. If villager have savings
  7. Ration card ownership
  8. Election card ownership
  9. Ration card classify
  10. Speak English or not
  11. If villager participate SHG or not
  12. Status

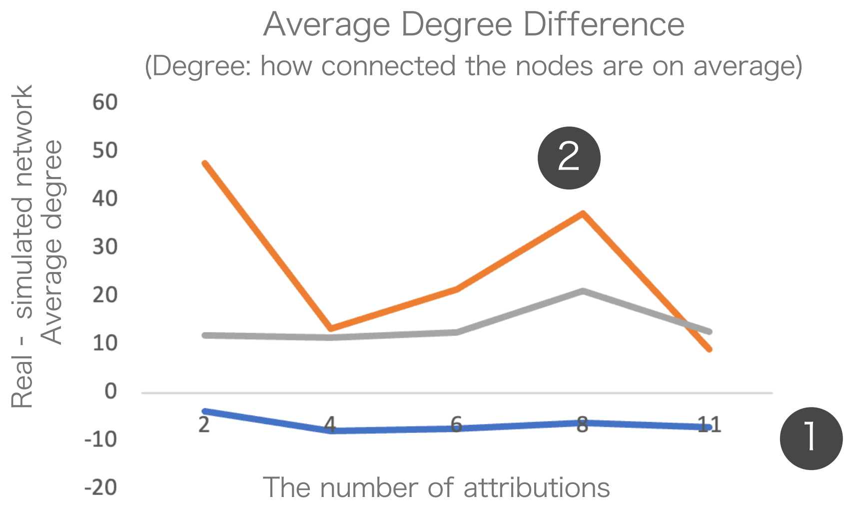
Results: Comparison of Simulated and Real Network Graph At Structural Features

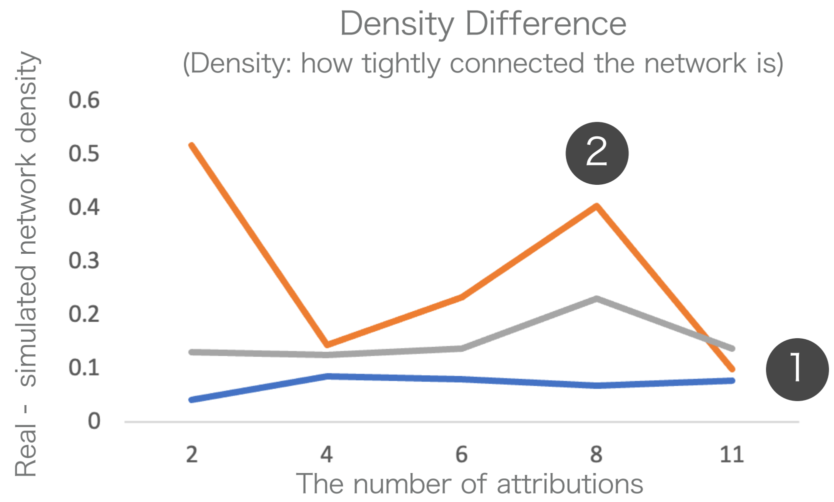
1
- Overall, the network generated by the global method is closest to the real network in terms of density and average degree,

2
- Local method demonstrates highest difference in 3 matrices, indicating poor performance

3
- However, the global method performs the worst when evaluated based on the local clustering coefficient.







Results: Comparison of Simulated and Real Network Graph At Edge Level

1
- The sequential method could achieved nearly 30% of precision, clearly outperforms others

2
- **Adding attribution is not helpful:** The precision, recall, and F1 scores showed no significant improvement as more attributes were added.

Generated structurally similar networks does not guarantee accuracy. Show the potential risk in relying LLM agents network prediction